

# Hormones and Their Receptors

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05\_Hormones\_and\_their\_receptors.pdf (25 slides) · Refs: Guyton & Hall 13th ed. (Unit II ch. 6); Ganong 25th ed. (Section I ch. 5)

**Legend:** ★ = high-yield · [added] = not on slides ·  = the lecturer's handwritten annotation. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

 **Bridge from Lecture 04 (Regulatory Systems):** Lecture 04 listed the endocrine glands, their hormones and the communication types, and ended by saying the nervous and endocrine systems are **interconnected**. This lecture shows **how** they are connected, what hormones **are chemically**, and **how a target cell detects them** (receptors and signalling).

## 1. Title / Objectives

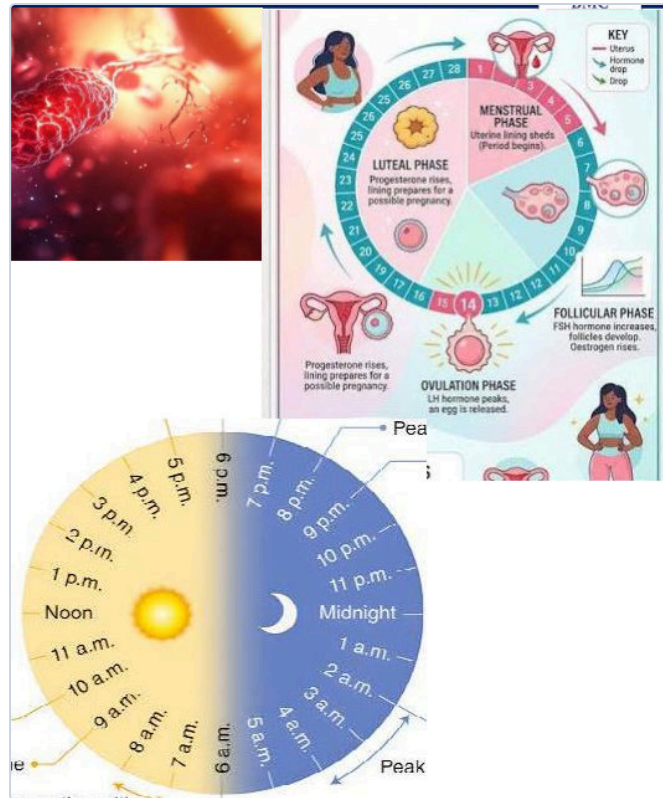
| Objective                                            | Likely tested as                                                            |
|------------------------------------------------------|-----------------------------------------------------------------------------|
| ★ Hormone <b>characteristics and chemical nature</b> | Properties list; steroid vs amine vs peptide examples                       |
| ★ Receptor <b>structure, distribution, ligands</b>   | Receptor location by hormone class; down/up-regulation; signalling pathways |

**Flow:** Properties of hormones → nervous-endocrine inter-relations → chemical nature → receptor locations → 2nd messengers → down-/up-regulation → intracellular signalling pathways → conclusion → case.

## 2. Main Content (Lecture Order)

### 2.1 Physiological Properties of Hormones (Slides 4-5)

- ★ Produced in **small amounts** ( $\mu\text{g}$  to  $\text{ng}$ )
- Rate of secretion** is set by the **body's needs**
- ★ **Superimposed rhythms:**
  - **Circadian** (daily), e.g., ★ **ACTH**
  - **Monthly**, e.g., **female gonadotropic hormones**
- Act on **specific target organs** (e.g., **sex hormones**) or on **many tissues** (e.g., **insulin**).  The lecturer wrote "synergise" here, meaning some hormones can also act together [interpretation].
- ★ Some hormones **antagonise** others (e.g., **insulin vs glucagon**)



Slide 5: Circadian and monthly (menstrual) hormonal rhythms

🔄 **Recap: Properties.** Tiny amounts ( $\mu\text{g}$ - $\text{ng}$ ), secreted on demand, with daily (ACTH) or monthly (gonadotropins) rhythms. Targeted (sex hormones) or widespread (insulin). Can antagonise (insulin/glucagon) or synergise.

🔗 **Link:** Hormone secretion is often switched on by the nervous system, and hormones in turn affect nerves: the interconnection promised in Lecture 04.

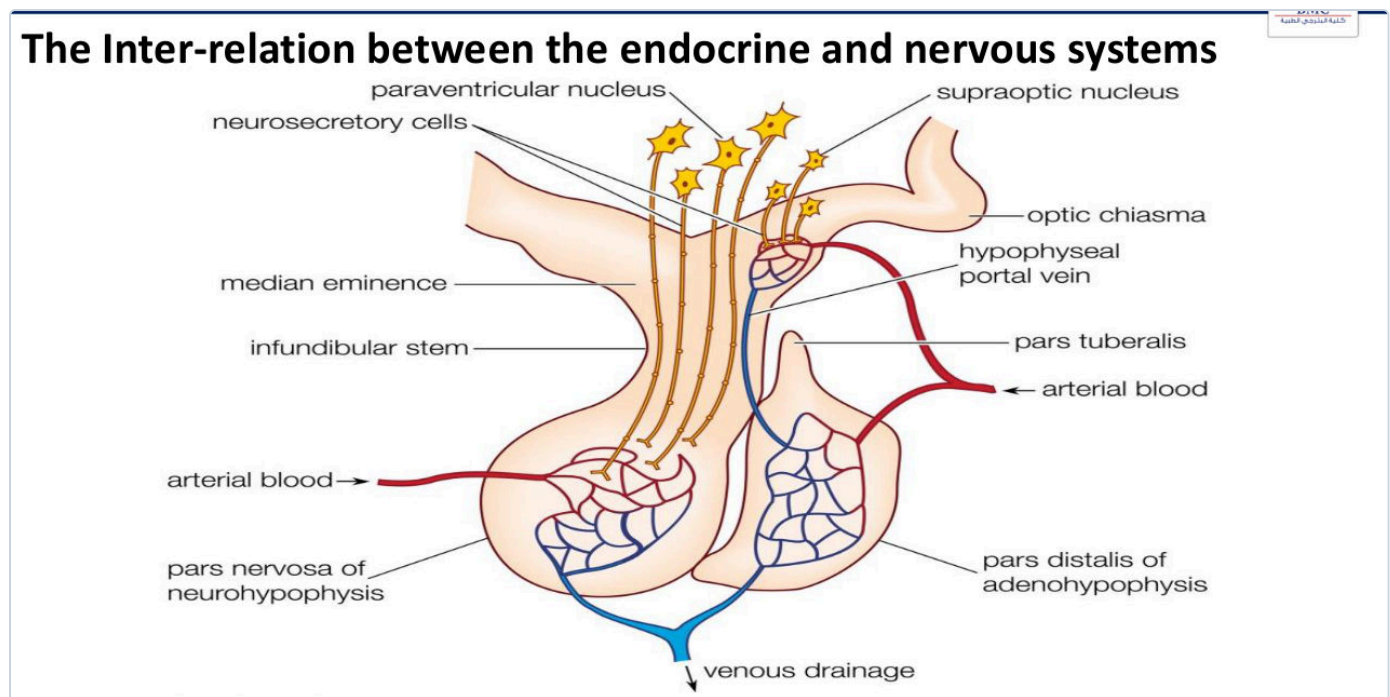
👉 **Up next: Nervous-endocrine inter-relations.**

## 2.2 Inter-relation Between the Nervous and Endocrine Systems (Slides 6-9)

### ★★ A. Nervous system → endocrine system

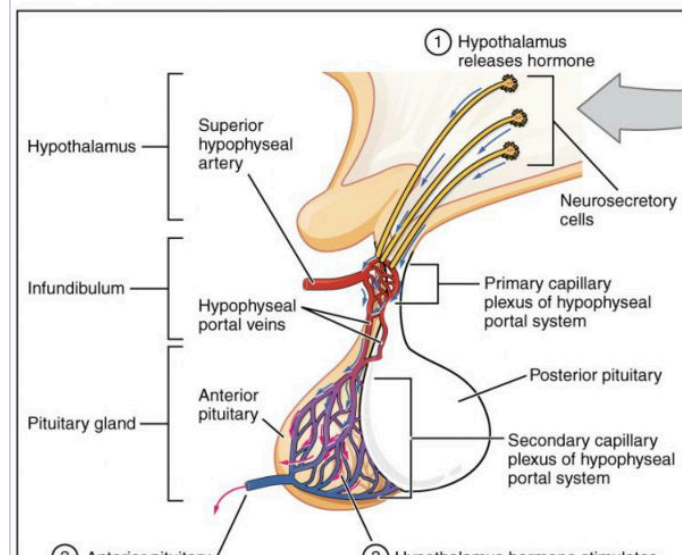
| #                                                                | Connection                                           | Between                                            | How                                                                                                                                                                    |
|------------------------------------------------------------------|------------------------------------------------------|----------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1                                                                | ★ <b>Hypothalamo-hypophyseal portal circulation</b>  | Hypothalamus → <b>anterior pituitary</b>           | Hypothalamic <b>releasing / inhibiting hormones</b> travel in the <b>portal blood</b> and alter anterior pituitary secretion (e.g., <b>GHRH</b> )                      |
| 2                                                                | ★ <b>Hypothalamo-hypophyseal tract</b> (nerve tract) | Hypothalamus → <b>posterior pituitary</b>          | ★ <b>ADH and oxytocin</b> are <b>formed in the hypothalamus</b> , then <b>stored and secreted</b> from nerve endings in the <b>posterior pituitary</b> (📝 "oxyt, ADH") |
| 3                                                                | <b>Direct neural connection</b>                      | Hypothalamus → <b>suprarenal (adrenal) medulla</b> | Controls <b>epinephrine and norepinephrine</b> release                                                                                                                 |
| (The slide numbers the third item "2." again, a numbering slip.) |                                                      |                                                    |                                                                                                                                                                        |

[added from the Slide 8 figure: oxytocin comes from the paraventricular nucleus and ADH from the supraoptic nucleus.]

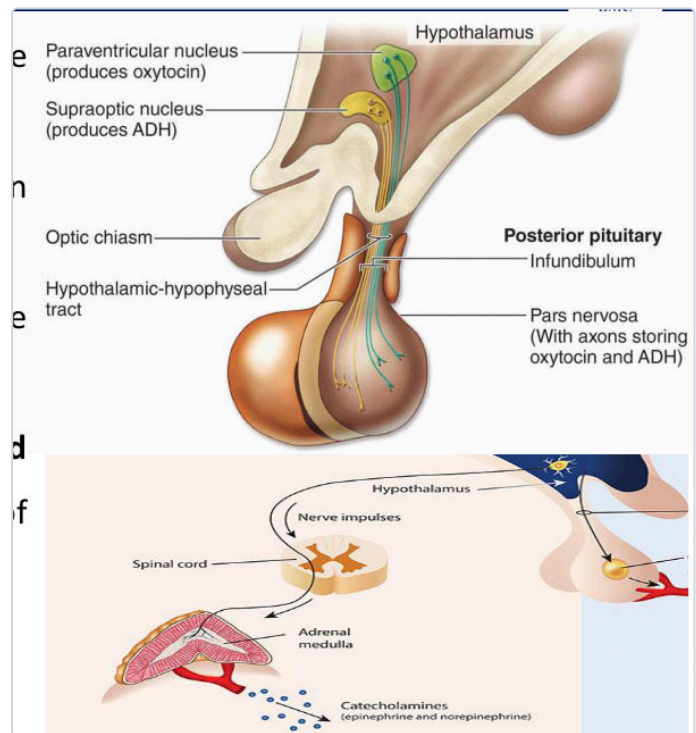


Slide 6: Hypothalamus, portal system and pituitary

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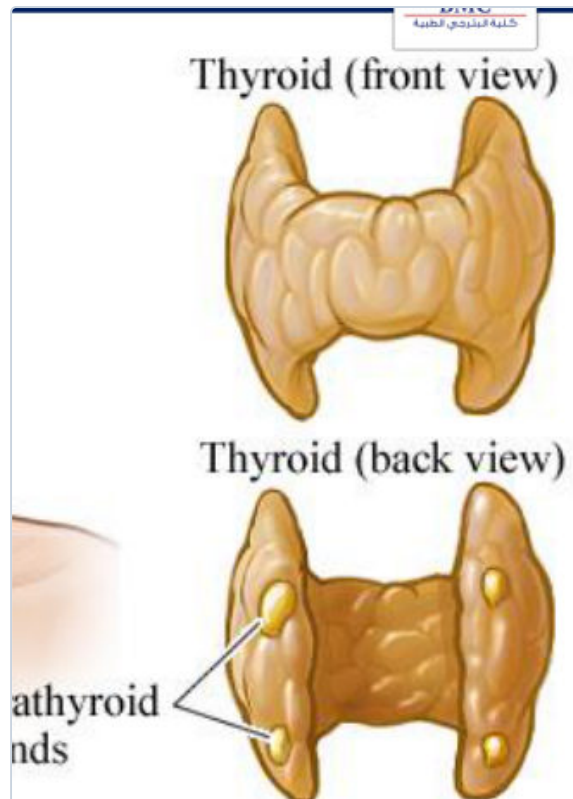
Slide 7: Hypothalamo-hypophyseal portal circulation (anterior pituitary)



Slide 8: Hypothalamo-hypophyseal tract (posterior pituitary) and neural control of the adrenal medulla

### ★ B. Endocrine system → nervous system

- **Parathyroids** control **blood  $\text{Ca}^{2+}$** , which affects **nerve excitability**
- **Thyroid hormones** affect **nervous-system excitability**



Slide 9: Thyroid and parathyroid glands

🔄 **Recap: Inter-relation.** Hypothalamus → anterior pituitary via portal **blood** (releasing/inhibiting hormones). Hypothalamus → posterior pituitary via **nerve tract** (ADH and oxytocin made in the hypothalamus). Hypothalamus → adrenal medulla by direct nerves. Back the other way: PTH ( $\text{Ca}^{2+}$ ) and thyroid hormones set nerve excitability.

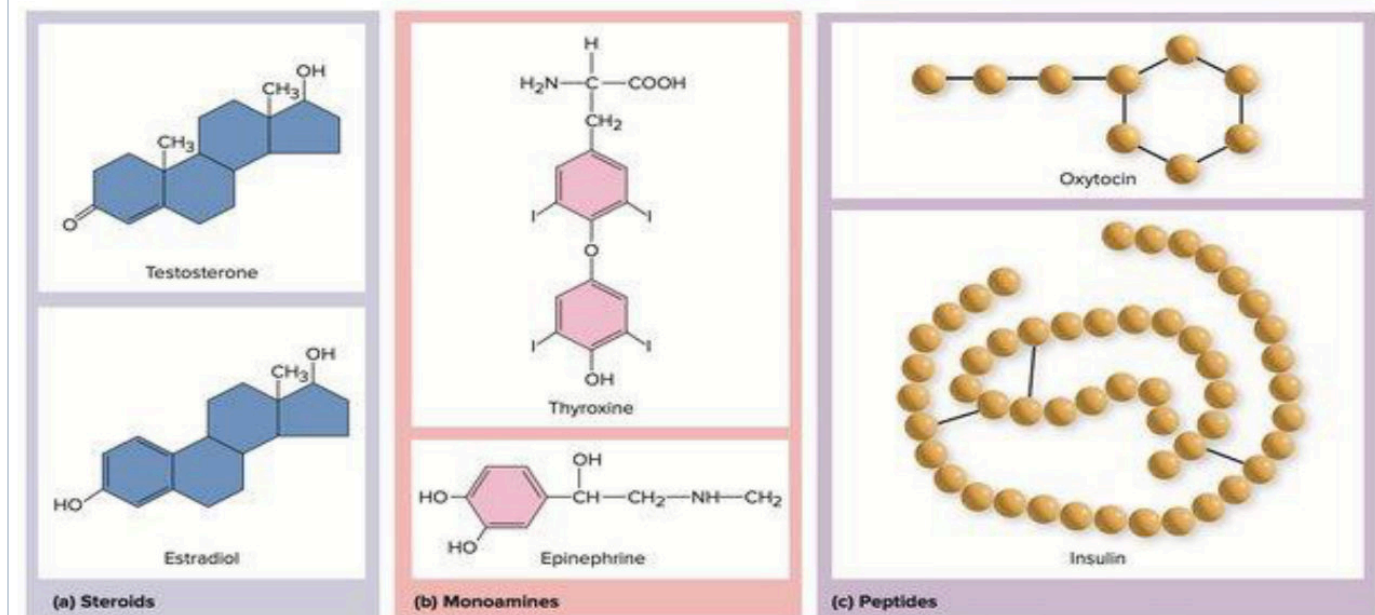
🔗 **Link:** Before looking at how cells detect hormones, classify them chemically, because chemistry decides where the receptor sits.

▶ **Up next: Chemical nature of hormones.**

## 2.3 Chemical Nature of Hormones (Slides 10-11)

| Class                                         | Source → hormones                                                                                                                              |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| ★ a) <b>Steroids</b>                          | <b>Adrenal cortex:</b> cortisol, aldosterone, androgens · <b>Placenta &amp; ovaries:</b> oestrogen, progesterone · <b>Testes:</b> testosterone |
| ★ b) <b>Amino-acid (tyrosine) derivatives</b> | <b>Thyroid:</b> thyroxine (T4), tri-iodothyronine (T3) · <b>Adrenal medulla:</b> dopamine, epinephrine, norepinephrine                         |
| ★ c) <b>Proteins &amp; polypeptides</b>       | <b>Pituitary</b> hormones · <b>Pancreas:</b> insulin, glucagon, somatostatin · <b>Parathyroid:</b> PTH · many others                           |

## ➤ Chemical Nature of Hormones:



Slide 10: Steroids (testosterone, estradiol), monoamines (thyroxine, epinephrine), peptides (oxytocin, insulin)

**Memory hook [added]: "Steroids come from the Cortex and Gonads; Tyrosine goes to Thyroid and adrenal medulla; everything else is Peptide."**

🔄 **Recap: Chemistry.** Steroids: adrenal cortex, gonads, placenta. Tyrosine derivatives: thyroid hormones and catecholamines. Proteins/peptides: pituitary, pancreas, PTH.

🔗 **Link:** A hormone does nothing until it binds its receptor.

👉 **Up next: Receptors.**

## 2.4 Receptors: Location & Second Messengers (Slides 12-14)

- ★ **First step** of hormone action: binding to **specific receptors** on the target cell
- ★ Cells **without the receptor** do not respond ("**silent**")

★★★★ **Receptor location:**

| # | Location                                 | Hormones                                                                            |
|---|------------------------------------------|-------------------------------------------------------------------------------------|
| 1 | ★ <b>In/on the cell membrane surface</b> | <b>Protein, peptide and catecholamine</b> hormones                                  |
| 2 | ★ <b>Cytoplasm</b>                       | <b>Steroid</b> hormones (primary receptors)                                         |
| 3 | ★ <b>Nucleus</b>                         | ★ <b>Thyroid hormones</b> , directly associated with <b>one or more chromosomes</b> |

## Hormone chemistry decides where the receptor is

### Protein / peptide

Pituitary hormones, insulin, glucagon, somatostatin, PTH

#### + Catecholamines

(tyrosine-derived: adrenaline, NA, dopamine)

#### Receptor: CELL MEMBRANE

water-soluble, can't cross membrane [added]

→ 2nd messengers (e.g., cAMP), channels, G-protein, enzymes

### Steroid

Adrenal cortex: cortisol, aldosterone, androgens  
Ovary/placenta: oestrogen, progesterone · Testis: testosterone

#### Receptor: CYTOPLASM

lipid-soluble, crosses membrane [added]

→ complex enters nucleus → transcription (new protein)

### Thyroid (T4, T3)

Tyrosine derivative, but behaves like a steroid (lipid-soluble) [added]

#### Receptor: NUCLEUS

bound to chromosomes

→ transcription modulation

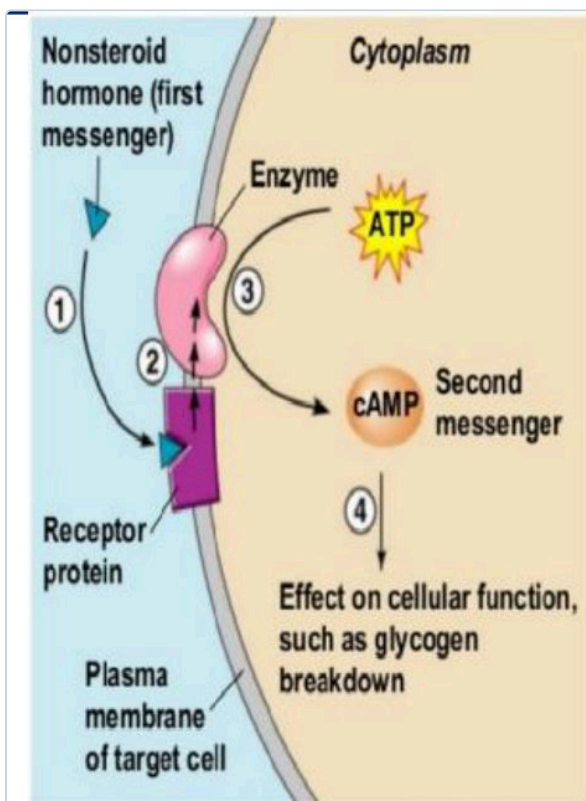
**Trap: tyrosine derivatives split. Catecholamines → membrane; thyroid hormones → nucleus.**

Cells without a receptor for the hormone are "silent" (no response).

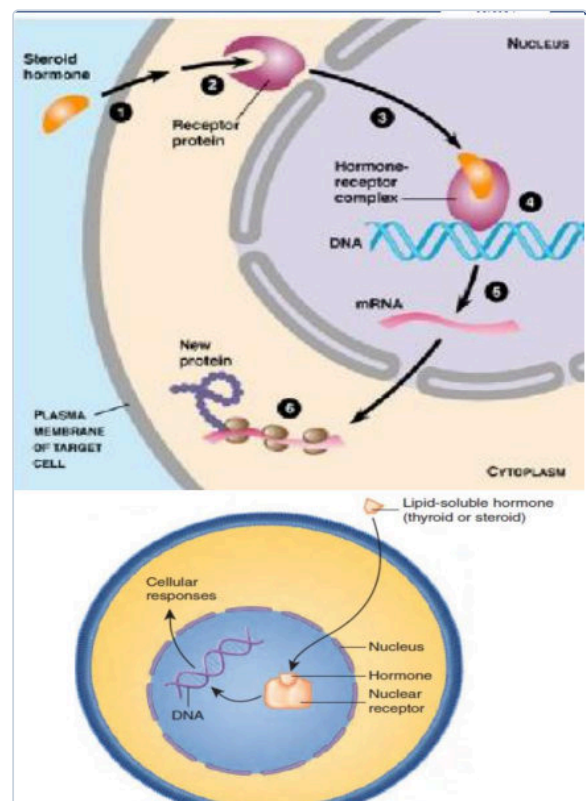
*Added diagram: hormone class → receptor location*

## ★ Messengers:

- Hormone = **"first messenger"**
- The substance produced **inside the cell** by hormone-receptor binding = ★ **"second messenger"** (e.g., **cAMP**, per the figure)
- ★ Second messengers can trigger reactions that **persist after the hormone has gone**



Slide 13: Membrane receptor → enzyme → cAMP (second messenger) → cell effect



Slide 14: Cytoplasmic (steroid) and nuclear (thyroid) receptors

🔗 **Recap: Location.** Membrane: peptides and catecholamines. Cytoplasm: steroids. Nucleus: thyroid. Hormone = 1st messenger; cAMP etc. = 2nd messenger, whose effects can outlast the hormone.

🔗 **Link:** Receptor number isn't fixed; it adjusts to hormone levels, which changes how sensitive the tissue is.

🔗 **Up next: Down- and up-regulation.**

## 2.5 Receptor Regulation (Slides 15-17)

★ **The number of receptors in a target cell is not constant.**

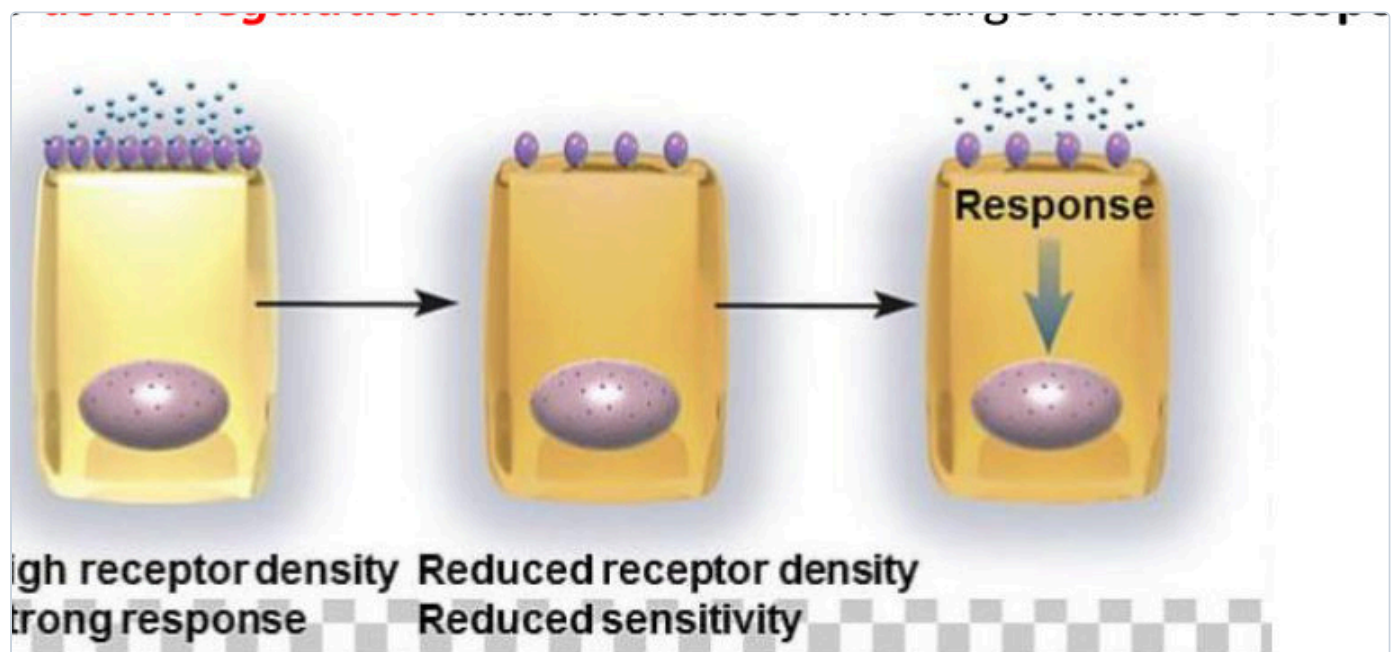
★★ **Down-regulation**

**Trigger:** ★ ↑ hormone concentration and ↑ binding → ↓ receptor number → ★ ↓ tissue responsiveness

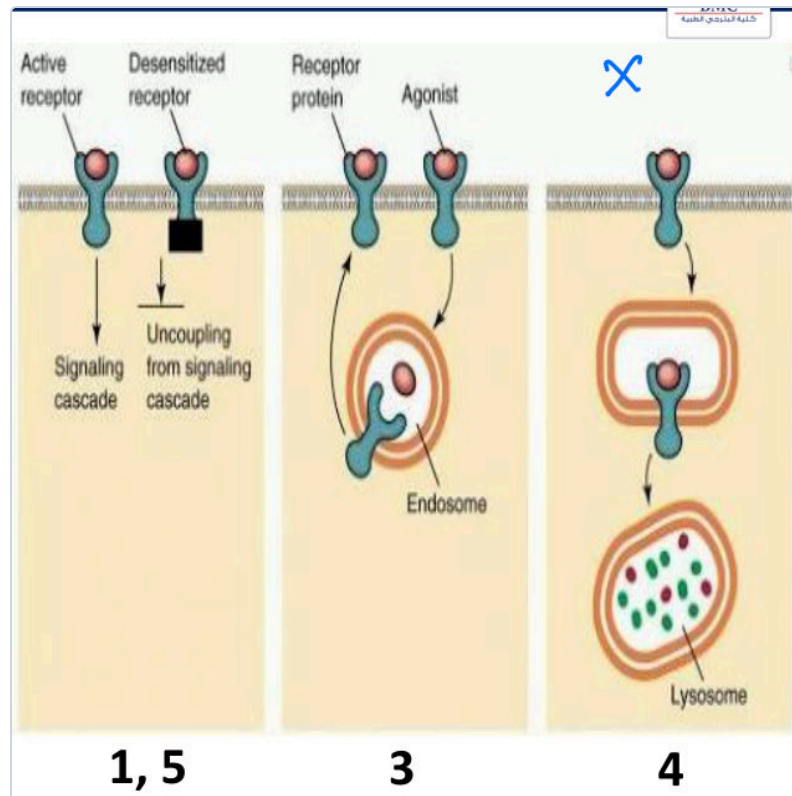
**Five mechanisms (Slide 16):**

1. **Inactivation** of some receptor molecules
2. **Decreased production** of receptors
3. **Temporary sequestration** of receptors inside the cell, away from the hormone
4. **Destruction by lysosomes** after internalisation
5. **Inactivation of intracellular signalling proteins**

Figure mapping (Slide 16): **1 & 5** = desensitised receptor uncoupled from the signalling cascade · **3** = receptor internalised in an endosome and recycled · **4** = receptor internalised and destroyed in a lysosome.



Slide 15: Down-regulation: fewer receptors, reduced sensitivity

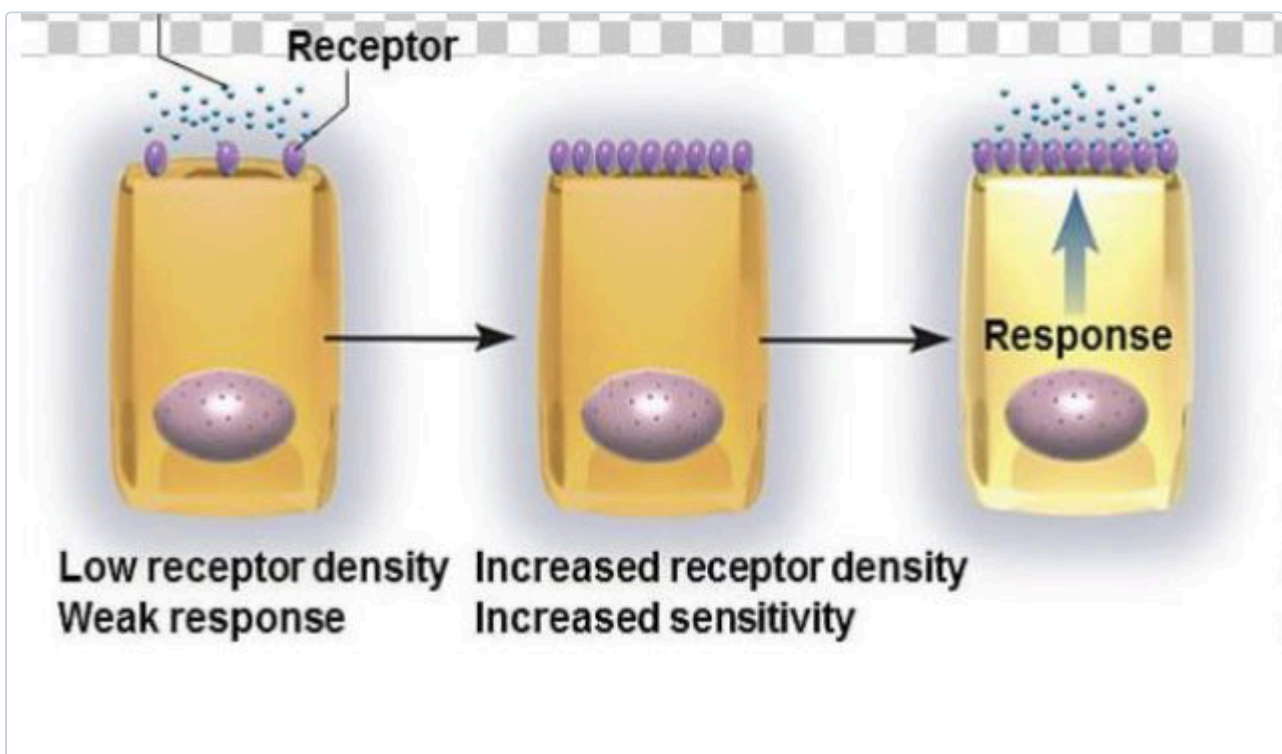


Slide 16: Down-regulation mechanisms (1 & 5 uncoupling, 3 endosome sequestration, 4 lysosomal destruction)

## ★★ Up-regulation

**Trigger:** ★ prolonged exposure to ↓ hormone → ↑ cell response

- Target tissue becomes **progressively more sensitive**
- Mechanisms: ★ ↑ **receptor expression** or ★ ↓ **rate of receptor internalisation**



Slide 17: Up-regulation: more receptors, increased sensitivity

🔁 **Recap: Regulation.** High hormone → down-regulation (inactivate, make fewer, sequester, destroy, or block signalling) → less sensitive. Low hormone for a long time → up-regulation (more receptors, less internalisation) → more sensitive.

🔗 **Link:** Once bound, the receptor has to convert the signal into a cellular effect. The lecture ends with the four ways it does that.

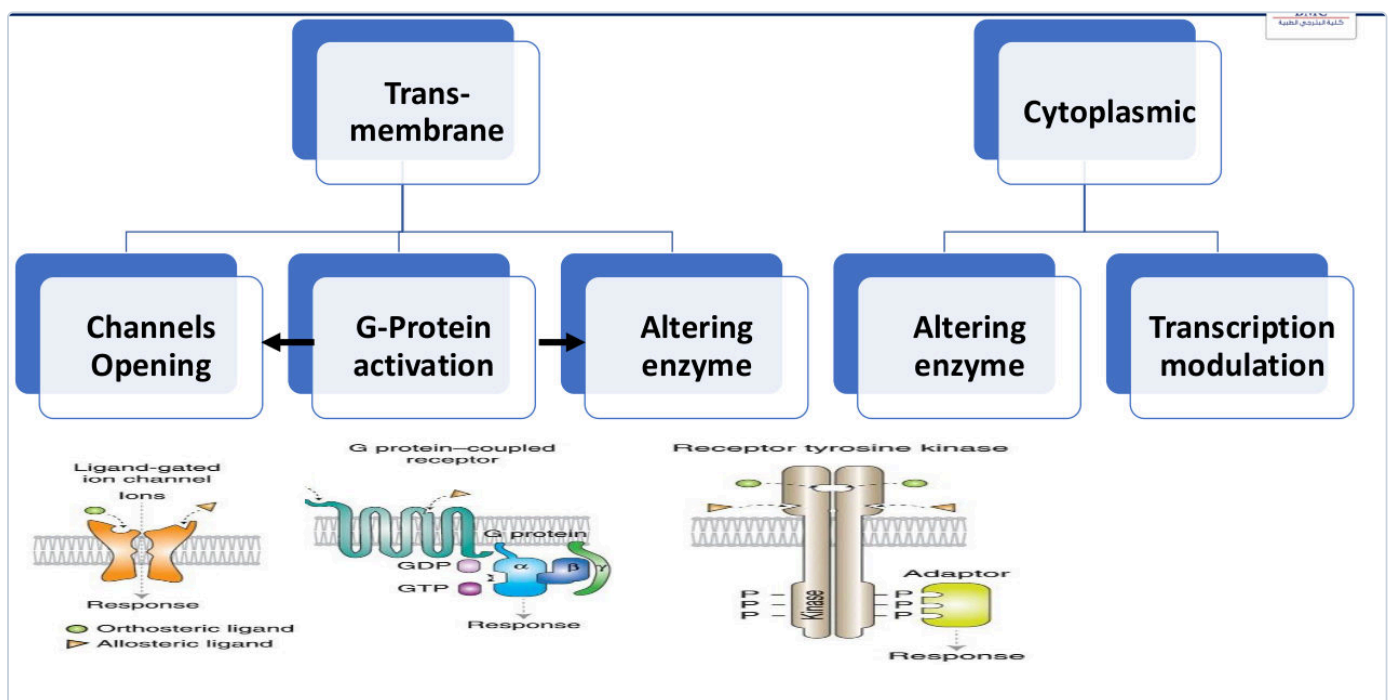
👉 **Up next: Intracellular signalling pathways.**

## 2.6 Intracellular Signalling Pathways (Slides 18-20)

★★★★ **Four mechanisms:**

|   | Mechanism                  | Receptor location            | Figure example                                                           |
|---|----------------------------|------------------------------|--------------------------------------------------------------------------|
| A | ★ Opening ion channels     | Transmembrane                | Ligand-gated ion channel                                                 |
| B | ★ G-protein activation     | Transmembrane                | G-protein-coupled receptor (GDP/GTP, $\alpha$ $\beta$ $\gamma$ subunits) |
| C | ★ Altering enzyme activity | Transmembrane or cytoplasmic | Receptor tyrosine kinase (+ adaptor)                                     |
| D | ★ Activating transcription | Cytoplasmic or nuclear       | Steroid/thyroid receptors                                                |

📖 On Slide 19, arrows from **G-protein activation** point to both **channel opening** and **altering enzymes**: a G-protein can act through either.



Slide 19: Signalling tree: transmembrane (channels, G-protein, enzymes) vs cytoplasmic (enzymes, transcription)

★ **What signalling pathways provide:**

- ★ **Amplification** (same effect, bigger) and **distribution** (different effects)
- ★ **Regulation and feedback control**

 **Recap: Signalling.** Transmembrane receptors open channels, activate G-proteins or alter enzymes. Cytoplasmic/nuclear receptors alter enzymes or transcription. Pathways amplify, distribute and allow feedback.

## 2.7 Conclusion (Slide 22)

- Hormones have **different characteristics**
- Hormones may be **protein, polypeptide, steroid, or tyrosine derivatives**
- Receptors may be **transmembrane, cytoplasmic, or nuclear**

## 3. High-Yield Comparison Tables

**Table A: Class → source → receptor → action**

|                         | Peptide/protein                   | Catecholamine       | Steroid                               | Thyroid                         |
|-------------------------|-----------------------------------|---------------------|---------------------------------------|---------------------------------|
| Chemistry               | Amino-acid chains                 | Tyrosine derivative | Cholesterol-derived<br><i>[added]</i> | Tyrosine derivative (iodinated) |
| Source                  | Pituitary, pancreas, parathyroid  | Adrenal medulla     | Adrenal cortex, gonads, placenta      | Thyroid                         |
| ★ Receptor              | <b>Membrane</b>                   | <b>Membrane</b>     | <b>Cytoplasm</b>                      | <b>Nucleus</b>                  |
| Action                  | 2nd messengers, channels, enzymes | 2nd messengers      | Transcription                         | Transcription                   |
| Onset<br><i>[added]</i> | Fast                              | Fast                | Slower                                | Slowest                         |

**Table B: Down- vs up-regulation**

|             | Down-regulation                                                                         | Up-regulation                   |
|-------------|-----------------------------------------------------------------------------------------|---------------------------------|
| Trigger     | ↑ hormone                                                                               | Prolonged ↓ hormone             |
| Receptors   | ↓                                                                                       | ↑                               |
| Sensitivity | ↓                                                                                       | ↑                               |
| Mechanisms  | Inactivation, ↓ production, sequestration, lysosomal destruction, ↓ signalling proteins | ↑ expression, ↓ internalisation |

**Table C: Hypothalamus → pituitary links**

|         | Anterior pituitary                         | Posterior pituitary                  |
|---------|--------------------------------------------|--------------------------------------|
| Link    | <b>Portal circulation (blood)</b>          | <b>Nerve tract (axons)</b>           |
| Carries | Releasing/inhibiting hormones (e.g., GHRH) | ADH, oxytocin (made in hypothalamus) |

**Table D: Signalling mechanisms (see §2.6)**

## 4. Rules, Exceptions, Patterns

- **Hormones act in µg-ng amounts.**

- **Every hormone action starts with receptor binding**; no receptor = silent cell.
- **Water-soluble (peptides, catecholamines) → membrane receptors. Lipid-soluble steroids → cytoplasm. Thyroid → nucleus.**
- **Tyrosine derivatives split**: catecholamines = membrane; thyroid = nucleus.
- **ADH and oxytocin are made in the hypothalamus, not the posterior pituitary** (which only stores and secretes them).
- **Anterior pituitary = blood (portal) link; posterior pituitary = nerve link.**
- **More hormone → fewer receptors; less hormone → more receptors.**
- **Second messengers can outlast the hormone.**
- **G-proteins can open channels or alter enzymes.**

## 5. Clinical Correlations *[added; the lecture has none]*

- **Type 2 diabetes / obesity**: chronically high insulin → insulin-receptor down-regulation → insulin resistance.
- **Tolerance to  $\beta$ -agonists** (e.g., salbutamol overuse) =  $\beta$ -receptor down-regulation.
- **Denervation hypersensitivity** = up-regulation after loss of the transmitter.
- **Beta-blocker withdrawal** → rebound tachycardia from up-regulated  $\beta$ -receptors.
- **Pituitary stalk section** cuts both links: anterior hormones fall (except prolactin) and ADH is lost (diabetes insipidus).
- **Hypocalcaemia** (e.g., after parathyroidectomy) → ↑ nerve excitability → tetany; the endocrine → nervous link.
- **Hyperthyroidism** → nervousness, tremor, brisk reflexes (↑ neural excitability); ties to the Thyroid 1 lecture.

## 6. Rapid Revision (Last-Day)

- Hormones:  **$\mu\text{g-ng}$** , secreted on demand, **circadian (ACTH) / monthly (gonadotropins)** rhythms, targeted (sex hormones) or wide (insulin), antagonism (insulin vs glucagon).
- Nervous → endocrine: **portal circulation → anterior pituitary** (releasing/inhibiting, e.g., GHRH); **hypothalamo-hypophyseal tract → posterior pituitary** (ADH, oxytocin made in hypothalamus); **direct nerves → adrenal medulla** (adrenaline, NA).
- Endocrine → nervous: **PTH ( $\text{Ca}^{2+}$ )** and **thyroid hormones** set excitability.
- Steroids: **adrenal cortex, ovaries, placenta, testes**. Tyrosine: **T4/T3, dopamine, adrenaline, NA**. Peptides: **pituitary, insulin, glucagon, somatostatin, PTH**.
- Receptors: **membrane** (peptide, catecholamine), **cytoplasm** (steroid), **nucleus** (thyroid).
- Hormone = 1st messenger; **2nd messenger** (cAMP) persists after the hormone.
- **Down-regulation** (↑ hormone): inactivation, ↓ production, sequestration, lysosomal destruction, ↓ signalling proteins.
- **Up-regulation** (prolonged ↓ hormone): ↑ expression, ↓ internalisation.
- Signalling: **channels, G-protein, enzymes** (transmembrane); **enzymes, transcription** (cytoplasmic/nuclear). Gives **amplification, distribution, regulation, feedback**.

## 7. Exam-Style Questions

**Q1 (MCQ, lecture case, Slide 23).** Where is the primary receptor for **cortisol** (a steroid)? 1) Mitochondria 2) Cell-membrane surface **3) Cytoplasm** 4) Nucleus, bound to chromosomes **Answer:**

**3.** Steroid receptors are cytoplasmic (Slide 14). Option 4 is the thyroid-hormone receptor; option 2 is for peptides and catecholamines. *[The slide doesn't mark an answer; this follows directly from Slide 14.]*

**Q2 (MCQ).** Which hormone's receptor is located in the nucleus? a) Insulin b) Epinephrine c) Aldosterone **d) Thyroxine Answer: d.**

**Q3 (MCQ).** ADH is: a) Synthesised in the posterior pituitary **b) Synthesised in the hypothalamus and released from the posterior pituitary** c) Carried to the pituitary by portal blood d) A steroid **Answer: b.** It travels down the hypothalamo-hypophyseal **tract** (axons), not the portal circulation.

**Q4 (MCQ).** Prolonged high hormone levels cause: **a) Down-regulation with reduced sensitivity** b) Up-regulation c) More receptor synthesis d) Reduced internalisation **Answer: a.**

**Q5 (Short answer).** List the four intracellular signalling mechanisms and where their receptors are. **Answer:** (A) ion channel opening, transmembrane; (B) G-protein activation, transmembrane; (C) altering enzyme activity, transmembrane or cytoplasmic; (D) transcription activation, cytoplasmic or nuclear.

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## 8. Exam Pitfalls & Traps

- 1. All tyrosine derivatives use membrane receptors.** No: **thyroid hormones are nuclear**; only catecholamines are membrane.
- 2. Steroid vs thyroid receptor:** steroid = **cytoplasm**; thyroid = **nucleus**.
- 3. ADH/oxytocin are made in the posterior pituitary.** No, they're made in the **hypothalamus** and only stored/released there.
- 4. Anterior pituitary is connected by nerves.** No, by **portal blood**; the posterior pituitary is the nerve connection.
- 5. Down-regulation = more receptors.** No: down = fewer, less sensitive; triggered by **high** hormone.
- 6. Second messengers stop when the hormone leaves.** Their effects **can persist**.
- 7. Enzyme-altering receptors are only membrane receptors.** They can be **transmembrane or cytoplasmic**.
- 8. ⚠ Slide numbering:** the slides list the adrenal-medulla connection as "2." again; it's the **third** nervous → endocrine link.
- 9. ⚠ Slide typos:** "placent and aovaries" = placenta and ovaries.